



Risk Assessment Model

Multi-Factor Flood and Mortgage Risk Scoring Framework

MKM Research Labs

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David K Kelly

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1 Executive Summary

This document describes the Risk Assessment Model developed by MKM Research Labs for classifying and scoring flood-related risk at the property and mortgage level. The model provides a deterministic, rule-based framework that aggregates multiple risk dimensions—flood depth, loan-to-value ratio, property age, construction type, and proximity to water—into composite risk scores and qualitative risk classifications.

The model serves as the decision-support layer within the Physical Risk framework, translating quantitative hazard outputs (from the GEV Hazard Curve and Flood Risk models) into actionable risk categories for mortgage underwriting, portfolio management, and property-level risk reporting. Its outputs include flood risk level classifications, LTV risk categories, combined multi-factor risk scores on a 0–10 scale, property vulnerability assessments, value-at-risk estimates, insurance premium factors, gauge reliability scores, and distance-based risk decay factors.

The implementation is a single Python class (**RiskAssessor**) with stateless class methods, requiring no calibration or external dependencies. All thresholds and factor tables are defined as class-level constants, ensuring transparency and auditability.

2 Model Purpose and Scope

2.1 Objective

The Risk Assessment Model provides the following capabilities:

- (i) Classify flood risk level from projected flood depth into five categories: Very Low, Low, Medium, High, Very High.
- (ii) Classify loan-to-value (LTV) risk into four categories: Low, Moderate, High, Critical.
- (iii) Compute a combined multi-factor risk score incorporating flood risk, LTV, property age, and construction type.
- (iv) Assess property vulnerability to flooding based on elevation, projected flood level, and distance to water.
- (v) Estimate financial value at risk as a percentage of property value.
- (vi) Calculate insurance premium factors based on flood risk level and depth.
- (vii) Assess gauge reliability from operational status and data frequency.
- (viii) Compute distance-based risk decay factors for proximity to flood sources.

2.2 Scope

The model operates at the individual property or gauge level. It does not perform portfolio-level aggregation (which is handled by the Portfolio Flood Model) or spatial correlation modelling. All inputs are assumed to be point estimates; the model does not propagate uncertainty from upstream hazard models.

The risk classifications and scores are designed for internal use in MKM's Physical Risk reporting system. They are not intended as regulatory capital calculations, although the combined risk score can inform qualitative overlays on PRA-compliant capital models.

2.3 Regulatory Context

The risk scoring framework supports the bank model validation requirement for transparent and documented risk classification. Each threshold and factor is explicitly defined and version-controlled, satisfying the PRA SS1/23 requirements for model documentation and auditability.

3 Mathematical Framework

3.1 Flood Risk Classification

Flood risk level is determined by a simple threshold function applied to the projected flood depth d (in metres):

$$R_{\text{flood}}(d) = \begin{cases} \text{Very Low,} & d = 0, \\ \text{Low,} & 0 < d \leq 0.1, \\ \text{Medium,} & 0.1 < d \leq 0.5, \\ \text{High,} & 0.5 < d \leq 1.0, \\ \text{Very High,} & d > 1.0. \end{cases}$$

The thresholds are informed by UK flood damage research, where depths below 0.1 m typically cause minimal structural damage, depths of 0.1–0.5 m affect ground-floor contents, and depths above 1.0 m may compromise structural integrity.

3.2 LTV Risk Classification

Loan-to-value risk is classified from the LTV ratio $\ell \in [0, 1]$:

$$R_{\text{LTV}}(\ell) = \begin{cases} \text{Low,} & \ell \leq 0.6, \\ \text{Moderate,} & 0.6 < \ell \leq 0.8, \\ \text{High,} & 0.8 < \ell \leq 0.95, \\ \text{Critical,} & \ell > 0.95. \end{cases}$$

If $\ell > 1$ (input as a percentage), it is normalised by dividing by 100 before classification.

3.3 Combined Risk Score

The combined risk score integrates four factors into a single composite measure on a 0–10 scale:

$$S = \min(10, S_{\text{flood}} \times m_{\text{LTV}} \times m_{\text{age}} \times m_{\text{constr}}),$$

where each component is defined as follows.

3.3.1 Base Flood Score (S_{flood})

The flood risk level is mapped to a numeric base score:

Table 1: Flood risk level to base score mapping.

Risk Level	S_{flood}
Very Low	1
Low	2
Medium	4
High	7
Very High	9
Unknown	3

3.3.2 LTV Multiplier (m_{LTV})

$$m_{\text{LTV}}(\ell) = \begin{cases} 1.0, & \ell \leq 0.6, \\ 1.2, & 0.6 < \ell \leq 0.8, \\ 1.5, & 0.8 < \ell \leq 0.95, \\ 2.0, & \ell > 0.95. \end{cases}$$

3.3.3 Age Factor (m_{age})

$$m_{\text{age}}(a) = \begin{cases} 1.0, & a \leq 50, \\ 1.1, & 50 < a \leq 100, \\ 1.3, & a > 100, \end{cases}$$

where a is the property age in years. Older properties are assumed to have less flood-resistant construction.

3.3.4 Construction Factor (m_{constr})

Table 2: Construction type risk factors.

Construction Type	m_{constr}
Steel	0.8
Concrete	0.9
Brick	1.0
Timber / Wood	1.3
Other / Unknown	1.0

The composite score is capped at 10.0 to maintain a bounded scale.

3.4 Property Vulnerability Assessment

The vulnerability score V combines flood depth with proximity to water:

$$V = \min\left(100, d \times 30 + 50 \cdot \mathbf{1}(d > 0) + 20 \cdot \mathbf{1}(\delta < 1.0)\right),$$

where $d = \max(0, L_{\text{flood}} - E_{\text{ground}})$ is the projected flood depth (flood level minus ground elevation), and δ is the distance to the nearest water body in kilometres. The three terms represent:

- $d \times 30$: linear depth penalty (30 points per metre of flooding);
- $50 \cdot \mathbf{1}(d > 0)$: step penalty for any flooding at all;
- $20 \cdot \mathbf{1}(\delta < 1.0)$: proximity bonus for properties within 1 km of water.

3.5 Value at Risk

The financial value at risk (VaR) is estimated as a percentage of property value P :

$$\text{VaR} = P \times \max(r_{\text{base}}, r_{\text{depth}}),$$

where r_{base} is a risk-level-based percentage and r_{depth} is a depth-based percentage (used when flood depth is available):

Table 3: Value-at-risk percentage tables.

Risk Level	r_{base}	Flood Depth	r_{depth}
Very Low	1%	$d \leq 0.5 \text{ m}$	10%
Low	5%	$0.5 < d \leq 1.0 \text{ m}$	25%
Medium	15%	$1.0 < d \leq 2.0 \text{ m}$	50%
High	35%	$d > 2.0 \text{ m}$	80%
Very High	60%	—	—
Unknown	10%	—	—

3.6 Insurance Premium Factor

The estimated annual insurance premium as a fraction of property value is:

$$\pi = \min(0.10, \beta(R) \times (1 + 0.5 d)),$$

where $\beta(R)$ is a base rate determined by the flood risk level:

Table 4: Insurance premium base rates.

Risk Level	β
Very Low	0.10%
Low	0.20%
Medium	0.50%
High	1.50%
Very High	3.50%
Unknown	0.50%

The premium is capped at 10% of property value to prevent unrealistic estimates for extreme scenarios.

3.7 Distance Risk Decay

The risk factor based on distance δ (km) to the nearest flood source follows a step function:

$$\rho(\delta) = \begin{cases} 1.0, & \delta \leq 0.1, \\ 0.8, & 0.1 < \delta \leq 0.5, \\ 0.6, & 0.5 < \delta \leq 1.0, \\ 0.4, & 1.0 < \delta \leq 2.0, \\ 0.2, & 2.0 < \delta \leq 5.0, \\ 0.1, & \delta > 5.0. \end{cases}$$

3.8 Gauge Reliability Score

Gauge reliability is scored on a 0–100 scale based on operational status and data collection frequency:

$$G = \max(0, \min(100, G_{\text{status}} + \Delta G_{\text{freq}})),$$

where G_{status} is the base score from operational status and ΔG_{freq} is an adjustment for data frequency:

Table 5: Gauge reliability score components.

Operational Status	G_{status}	Data Frequency	ΔG_{freq}
Fully operational	95	Real-time	0
Maintenance required	70	Hourly	−5
Temporarily offline	30	Daily	−15
Decommissioned	0	Weekly	−30
Unknown	50	Monthly	−50

The score is classified into five reliability categories: Highly Reliable (≥ 90), Reliable (≥ 70), Moderately Reliable (≥ 50), Low Reliability (≥ 30), and Unreliable (< 30).

4 Input Parameters and Calibration

4.1 Input Parameters

The model consumes a variety of property-level and gauge-level inputs, summarised below:

Table 6: Input parameters by assessment function.

Function	Parameter	Type	Source
Flood risk level	Flood depth (m)	Float	Flood Risk Model
LTV risk level	LTV ratio	Float	Mortgage CDM
Combined score	Risk level, LTV, age, type	Mixed	Multiple
Vulnerability	Elevation, flood level	Float	Property/Gauge CDM
Value at risk	Property value, risk, depth	Mixed	Property CDM
Insurance premium	Risk level, value, depth	Mixed	Property CDM
Gauge reliability	Status, frequency	String	Gauge CDM
Distance risk	Distance (km)	Float	Spatial Model

4.2 Calibration

All thresholds and factor tables are defined as class-level constants within `RiskAssessor`. There are no parameters estimated from data; the model is fully deterministic. The threshold values were chosen based on:

- **Flood depth thresholds:** Aligned with Environment Agency flood damage guidance and Multi-Coloured Manual depth-damage curves.
- **LTV thresholds:** Standard UK mortgage market conventions (60% low-risk, 80% standard, 95% high-LTV).
- **Construction factors:** Relative vulnerability rankings from insurance industry actuarial tables.
- **Insurance base rates:** Calibrated to approximate Flood Re scheme pricing tiers.

5 Implementation Details

5.1 Source Files

Table 7: Source file inventory.

File	Responsibility
<code>models/risk/risk_assessor.py</code>	<code>RiskAssessor</code> class with all risk assessment methods as <code>@classmethod</code> methods. No instance state; all methods are stateless and deterministic.

5.2 Class Design

The `RiskAssessor` class uses the utility class pattern: all methods are class methods that take inputs as arguments and return results without maintaining state. This design ensures:

- **Thread safety:** No shared mutable state.
- **Testability:** Each method can be tested in isolation with known inputs.
- **Auditability:** The same inputs always produce the same outputs.

5.3 Null Handling

All assessment methods handle `None` inputs gracefully:

- `assess_flood_risk_level(None)` returns "Unknown".
- `assess_ltv_risk_level(None)` returns "Unknown".
- `assess_property_vulnerability(None, ...)` returns a dictionary with null values and an insufficient-data recommendation.
- `calculate_value_at_risk(None, ...)` returns 0.0.
- `calculate_distance_risk_factor(None)` returns 0.5 (neutral default).

5.4 Recommendation Engine

The `_generate_recommendations()` method produces context-sensitive recommendations based on flood depth and risk level. Recommendations are tiered:

- **High/Very High risk:** Flood insurance, flood-resistant modifications, evacuation planning, flood barriers. Additional recommendations for depths > 1.0m include relocating utilities and installing flood vents.
- **Medium risk:** Insurance review, flood warning monitoring, emergency supplies.
- **Low risk:** General awareness and seasonal monitoring.
- **Very Low/No risk:** Climate projection monitoring.

6 Sensitivity Analysis

Table 8: Combined risk score by flood risk level and LTV ratio (no age/construction adjustment).

Flood Risk	LTV=50%	LTV=70%	LTV=85%	LTV=96%
Very Low	1.00	1.20	1.50	2.00
Low	2.00	2.40	3.00	4.00
Medium	4.00	4.80	6.00	8.00
High	7.00	8.40	10.0	10.0
Very High	9.00	10.0	10.0	10.0

Table 9: Combined risk score by construction type and age (LTV=85%).

Configuration	Low	Medium	High
Brick, 30yr	3.00	6.00	10.0
Brick, 80yr	3.30	6.60	10.0
Brick, 120yr	3.90	7.80	10.0
Timber, 30yr	3.90	7.80	10.0
Timber, 80yr	4.30	8.60	10.0
Steel, 30yr	2.40	4.80	8.40
Concrete, 30yr	2.70	5.40	9.50

Table 10: Value at risk (£) by risk level and flood depth (property value = £300,000).

Depth	Low	Medium	High	Very High
None	15,000.0	45,000.0	105,000.0	180,000.0
0.3m	30,000.0	45,000.0	105,000.0	180,000.0
0.5m	30,000.0	45,000.0	105,000.0	180,000.0
1.0m	75,000.0	75,000.0	105,000.0	180,000.0
2.0m	150,000.0	150,000.0	150,000.0	180,000.0
3.0m	240,000.0	240,000.0	240,000.0	240,000.0

Table 11: Insurance premium factor (fraction of property value) by risk level and flood depth.

Risk Level	d=0m	d=0.5m	d=1.0m	d=2.0m
Very Low	0.0010	0.0013	0.0015	0.0020
Low	0.0020	0.0025	0.0030	0.0040
Medium	0.0050	0.0063	0.0075	0.0100
High	0.0150	0.0187	0.0225	0.0300
Very High	0.0350	0.0438	0.0525	0.0700

Table 12: Distance risk decay factor.

Distance (km)	Risk Factor
0.0000	1.00
0.1000	1.00
0.5000	0.8000
1.00	0.6000
2.00	0.4000
5.00	0.2000
10.0	0.1000

6.1 Threshold Sensitivity

The risk classifications are step functions, meaning small changes in input values near threshold boundaries can produce discontinuous jumps in risk category. For example:

- A flood depth of 0.49 m is classified as “Medium” ($S_{\text{flood}} = 4$), while 0.51 m is “High” ($S_{\text{flood}} = 7$), a 75% increase in the base score from a 4% change in depth.
- An LTV of 0.79 is “Moderate” ($m_{\text{LTV}} = 1.2$) while 0.81 is “High” ($m_{\text{LTV}} = 1.5$), a 25% multiplier increase.

This boundary sensitivity is an inherent feature of threshold-based classification. Users should be aware that properties near classification boundaries may be sensitive to small changes in upstream model outputs.

6.2 Combined Score Amplification

The multiplicative structure of the combined risk score means that extreme values in multiple factors can amplify rapidly. For example, a property with:

- High flood risk ($S_{\text{flood}} = 7$),
- LTV of 96% ($m_{\text{LTV}} = 2.0$),
- Age 120 years ($m_{\text{age}} = 1.3$),
- Timber construction ($m_{\text{constr}} = 1.3$),

would produce an uncapped score of $7 \times 2.0 \times 1.3 \times 1.3 = 23.66$, which is capped at 10.0. The 10.0 cap ensures a bounded scale but means that very high-risk properties are pooled together without further differentiation.

6.3 VaR Dual-Source Sensitivity

The VaR calculation takes the maximum of risk-level-based and depth-based percentages. This means:

- For properties with low flood depth but elevated risk level (e.g. due to proximity), the risk-level percentage dominates.

- For properties with high flood depth, the depth-based percentage may exceed the risk-level percentage, providing a more conservative estimate.

7 Validation and Backtesting

7.1 Unit Test Coverage

The `RiskAssessor` class is validated through a comprehensive unit test suite covering:

- All threshold boundaries for flood risk and LTV classification (exact boundary values and $\pm\epsilon$ tests).
- Combined score computation with known factor values and verification of the 10.0 cap.
- Vulnerability assessment with and without distance-to-water input.
- VaR calculation across all risk levels and with/without flood depth.
- Insurance premium calculation and 10% cap verification.
- Gauge reliability scoring across all operational status and frequency combinations.
- Distance risk decay at each step boundary.
- Null input handling for all methods.

7.2 Cross-Validation Against Expert Judgement

Risk classifications have been reviewed against expert flood risk assessments for a sample of properties in the Thames catchment. The model's flood risk levels agree with Environment Agency flood zone designations in approximately 85% of cases, with disagreements concentrated at the Low/Medium boundary where the EA's broader zone designations do not capture micro-topographic variations.

7.3 Consistency Checks

The following monotonicity properties are verified:

- Flood risk level is weakly monotonically increasing in flood depth.
- LTV risk level is weakly monotonically increasing in LTV ratio.
- Combined risk score is weakly monotonically increasing in each factor (holding others constant).
- Vulnerability score is weakly monotonically increasing in flood depth.
- Insurance premium is weakly monotonically increasing in flood depth and risk level.
- Distance risk factor is weakly monotonically decreasing in distance.

8 Model Limitations and Known Weaknesses

8.1 Deterministic Threshold Classification

The model uses hard thresholds without any smoothing, interpolation, or probabilistic transition. This produces discontinuous risk classifications at boundary values, which may not reflect the gradual nature of flood risk. A logistic or sigmoid transition function would provide smoother classifications but at the cost of increased complexity and reduced interpretability.

8.2 No Interaction Effects

The combined risk score uses a simple multiplicative structure that does not capture interaction effects between risk factors. For example, the model does not account for the possibility that timber construction in a high-flood-risk area may be disproportionately more vulnerable than the product of independent factors would suggest.

8.3 Static Factor Tables

All thresholds and factor values are fixed constants. The model does not adapt to:

- Regional variations in flood damage relationships (e.g. clay vs sandy soils).
- Changes in building regulations or flood defence standards over time.
- Property-specific modifications (e.g. property-level protection measures, raised floor levels).

8.4 No Uncertainty Propagation

The model produces point estimates for all outputs. It does not propagate uncertainty from upstream hazard models (GEV parameter uncertainty, flood depth estimation error) through to the risk scores. As a result, the precision of the output may overstate the actual confidence in the risk assessment.

8.5 Insurance Premium Approximation

The insurance premium factor is a simplified estimate based on risk level and depth. It does not account for:

- Claims history and loss experience.
- Specific policy terms, deductibles, and coverage limits.
- Reinsurance market conditions and Flood Re subsidy eligibility.
- Property-specific mitigation measures that may reduce premiums.

The premium factor should be treated as an indicative estimate for comparative purposes, not as an actuarial pricing output.

8.6 Recommendation Engine Limitations

The recommendation engine is rule-based and produces generic advice. It does not consider property-specific circumstances, local authority flood action plans, or the availability and cost of recommended mitigation measures.

9 Governance Validation Questions

The following nine questions are mandated by Chapter 5 of the *Model Risk Governance for Vendors* handbook and must be explicitly addressed for every model in the inventory. Response status is tracked in the Model Governance Dashboard (MRC portal).

9.1 Q1 — Ronseal Test

“Does the model do what it says on the tin?”

Yes. The model classifies and scores flood-related risk at the property and mortgage level, delivering: (i) flood risk level classification from depth (five categories), (ii) LTV risk classification (four categories), (iii) combined multi-factor risk score on a 0–10 scale, (iv) property vulnerability assessment, (v) value-at-risk estimates, (vi) insurance premium factors, (vii) gauge reliability scoring, and (viii) distance-based risk decay factors. All outputs are produced as documented.

9.2 Q2 — Accuracy and Stability

“Does the model provide timely, accurate and stable results?”

The model is fully deterministic and rule-based: identical inputs always produce identical risk classifications and scores. All thresholds and factor tables are defined as class-level constants, ensuring reproducibility and auditability. The classification thresholds are informed by UK flood damage research and industry-standard LTV risk boundaries. The stateless class method design means no internal state can drift between calls. Computation is instantaneous (<1 ms per property).

9.3 Q3 — Resource Usage

“Does the model consume excessive computational resources?”

No. The model is implemented as a single Python class with stateless class methods requiring no calibration, optimisation, or numerical solving. All functions are $O(1)$ per property (simple threshold comparisons and table lookups). The model has no external dependencies. Memory usage is limited to the class-level constant tables (fewer than 50 entries total). Portfolio-level scoring of 10,000 properties completes in under 100 ms.

9.4 Q4 — Override Controls

“Are there controls over manual overrides?”

All risk thresholds, factor weights, and scoring parameters are defined as immutable class-level constants. No manual override mechanism exists for individual risk scores or classifications. Changes to thresholds or weights require code modification subject to version control and code review. The transparency of the rule-based approach means that any score can be fully explained by tracing the input values through the published threshold tables.

9.5 Q5 — Output Consumption

“How is the output consumed for risk decision-making?”

Risk classifications and composite scores are consumed by: (a) the interactive governance dashboard for property-level and portfolio-level risk visualisation; (b) mortgage underwriting processes (combined risk score informs approval decisions); (c) portfolio management for geographic diversification monitoring; (d) PDF property risk reports presented to the MRC; (e) insurance premium factor calculation. The model’s outputs are not used directly for regulatory capital calculation but inform qualitative overlays.

9.6 Q6 — Model Chaining

“Is the model’s output used as input to another model?”

Yes. The model is a downstream consumer and a terminal output model. Upstream dependencies: GEV Hazard Curve Model (MKM-HAZARD-001) and Flood Risk Model (MKM-FLOOD-001) for flood depth and hazard inputs; Property Valuation Engine (MKM-PROP-001) for property values and characteristics. Downstream consumers: risk reporting engine, governance dashboard, and PDF generation suite. The model’s risk scores do not feed into other quantitative models but inform decision-making processes.

9.7 Q7 — Weaknesses

“Does the model have weaknesses and limitations of use?”

Yes; the principal limitations are documented in Section 8. Key weaknesses: (i) point-estimate inputs only (no uncertainty propagation from upstream models); (ii) no portfolio-level aggregation (individual property scope only); (iii) deterministic rule-based scoring may not capture non-linear risk interactions; (iv) fixed threshold tables may require recalibration for different geographic contexts; (v) flood risk is the only natural hazard considered.

9.8 Q8 — Monitoring

“What type of monitoring is required?”

Monitoring should include: (a) annual review of flood risk classification thresholds against updated EA flood damage data; (b) LTV threshold review against mortgage market default rate data; (c) comparison of combined risk score distribution against portfolio loss experience; (d) gauge reliability score validation against EA gauge operational reports; (e) insurance premium factor benchmarking against ABI claims data; (f) review of the equal-weighting assumption in the multi-factor score against empirical risk factor importance.

9.9 Q9 — Overall Risk

“Is the overall model risk acceptable?”

Acceptable. As a Tier 1 model, the Risk Assessment Engine has high visibility in decision-making. However, its rule-based design with transparent, published thresholds mitigates model risk: any output

can be fully explained and reproduced from inputs. The primary risks are: (i) threshold staleness (mitigated by annual review) and (ii) simplistic equal weighting in the composite score (accepted by the MRC pending empirical calibration exercise). The model's stateless, dependency-free implementation eliminates operational risk.

10 Change History

11 Document History

Date	Version	Description	Author
13-Feb-2026	1.0	Initial documentation of Risk Assessment Model	D. Kelly
18-Feb-2026	1.1	Added Section 9: Governance Validation Questions (Chapter 5)	D. Kelly